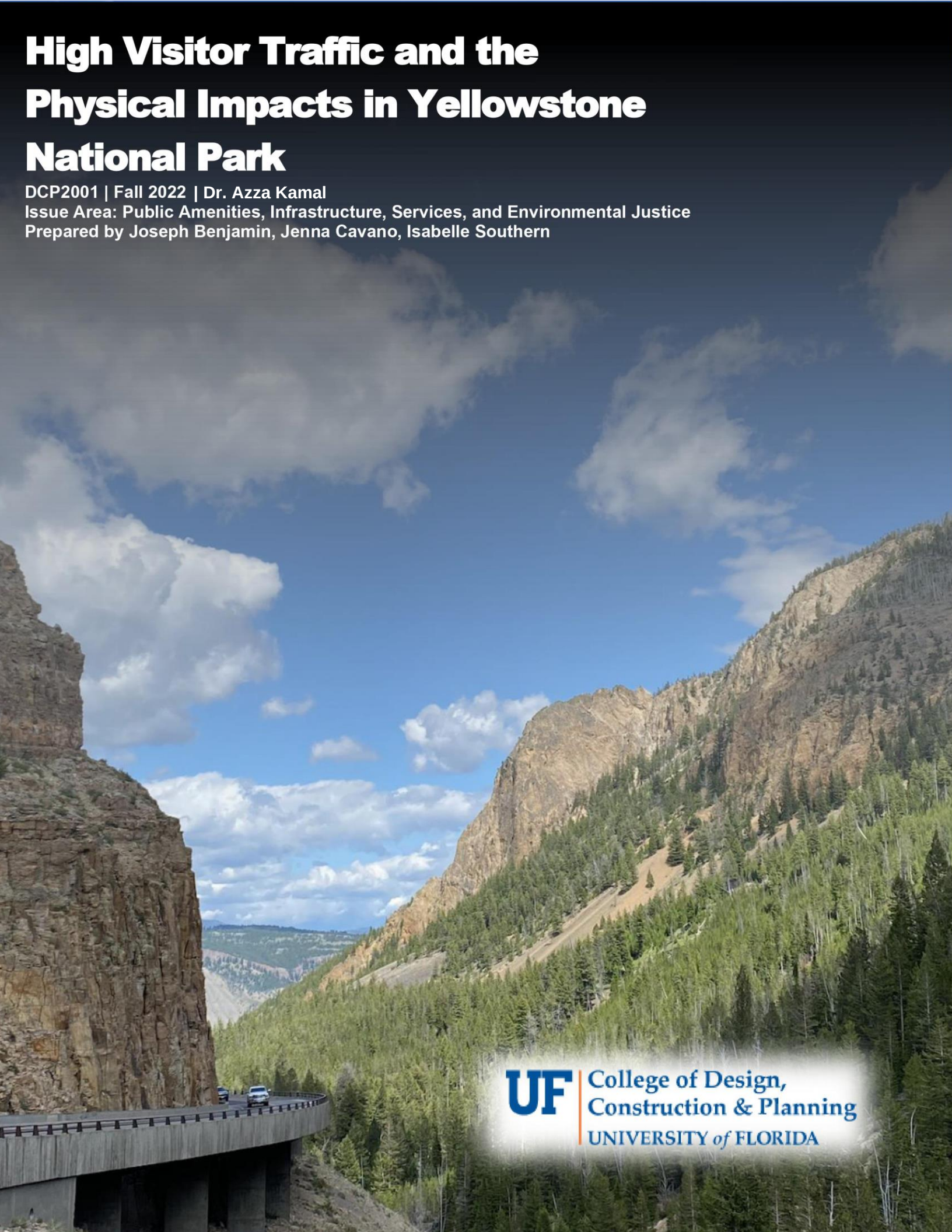
High Visitor Traffic and the Physical Impacts in Yellowstone National Park

DCP2001 | Fall 2022 | Dr. Azza Kamal
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Defining the Problem

Established in 1872, Yellowstone National Park has grown to host millions of visitors annually. The physical landscape and ecology of the park has been altered by built development and high visitor traffic in the park's wild environment. Open back-country hiking, camping, facilities for tourists, and impervious surface development for vehicle traffic are a few examples of this human presence. This report will be conducted as a mixed-methods analysis of how human-built features impact the physical attributes of Yellowstone's natural ecosystem.

Research Questions

This study will be framed by two questions that concern chronological spatial change in the Area of Interest, Yellowstone National Park. Based on the spatial and tabular data reported by federal park and recreation agencies, how does trail and roadway development in Yellowstone National Park through time correlate with increasing tourism population densities? To what extent has built development, accounting for the growing number of visitors, denaturalized previously uninhabited land in the park?

Background

Yellowstone was established in 1872 as the first National Park in the United States (NPS), however there is evidence of human activity in the modern park boundaries for at least 11,000 years. Oral histories of the Kiowa place their ancestors in the area until the 1700s. Yellowstone National Park covers about 3,468 square miles, hosts nine visitor centers, and has twelve campgrounds. The Organic Act established the National Park Service (NPS) for the

purposes of conservation and enjoyment (Organic Act, 1916). These established purposes can come into conflict with one another, complicating best management practices (Tranel and Hall, 2003). National Parks, especially the oldest and most iconic, are viewed as sacred spaces central to the quasi-religious nature of American identity (Ross-Bryant, 2005). Internationally, this has been identified as one of the biggest challenges that national parks face (Ferretti-Gallon et al., 2021). However, there are times when decisions are made to limit human access for the preservation of non-human life and landforms. There are also instances of changing the landscape or displacing wildlife and vegetation for the purposes of human recreation.

Public appeal toward wilderness and the great outdoors has attracted tourists into Yellowstone National Park since its creation. Visitor traffic to U.S. National Parks has greatly increased from the parks' humble beginnings (NPS, 2021). The COVID-19 pandemic disrupted record high visitation numbers, but numbers are beginning to rise again as travel restrictions are lifted and a sense of new-normalcy is established. In particular, the ten years with the highest levels of tourism for Yellowstone have all been in the last two decades, as can be seen in the Figure below (NPS Stats, 2022).

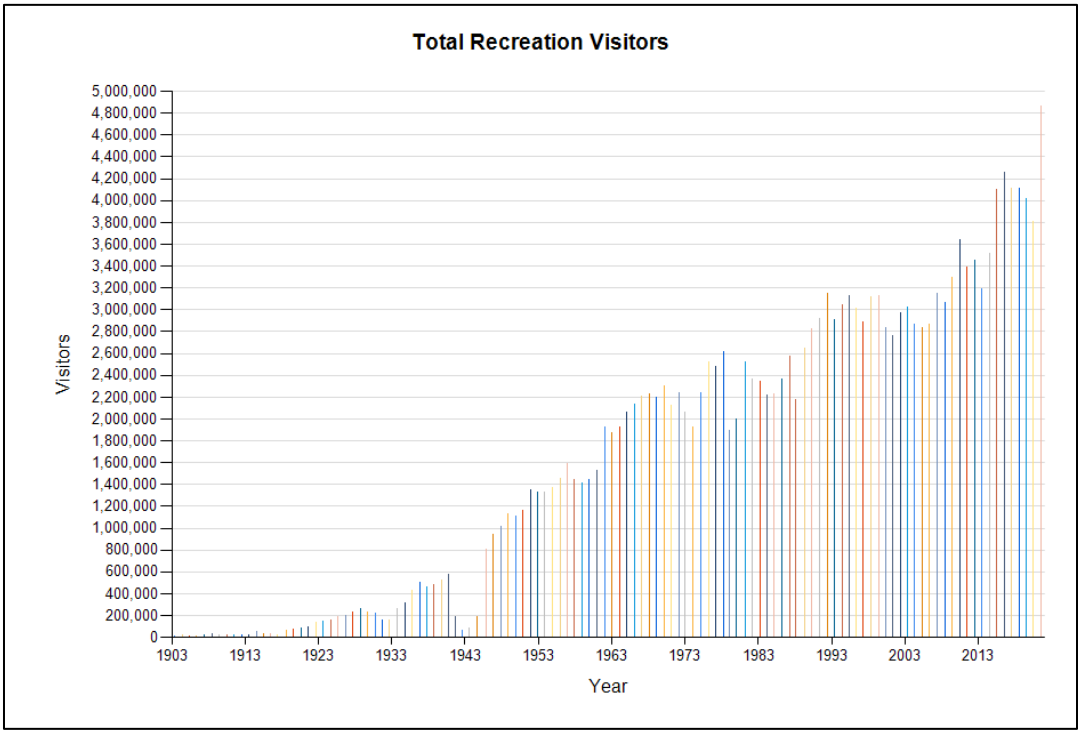


Figure 1: Annual Park Recreation Visitation Graph (1904 – 2021)

Over time, the physical landscape and ecology of the park has greatly changed. This is in part because of high visitor traffic. Also associated with these changes are park management decisions, such as the selective protection and hunting of species, backcountry foot traffic, impermeable roadway development, and amenity construction altering open spaces. Progressive increases in tourism through time have required development of built features to accommodate large numbers of park visitors at once.

Despite Yellowstone's wild characterization, the man-made infrastructure can be found throughout the park. Five entryways to the park connect to the Grand Loop, the main roadway connecting visitors to the park's attractions (O'Brien, 1966). These roads, along with secondary roads that lead to the nearly two thousand buildings throughout the park (Secret Yellowstone). A

rich trail system has enabled guests to be immersed in nature but are prone to degradation and soil erosion over time (Marion 2006). This project hypothesizes that the physical layout of Yellowstone National Park shows these spatial changes, where the increased ratio of built versus natural land area is represented geospatially.

Historic mapping and construction reports shape much of the analysis of built environment development in Yellowstone. Native Americans provide the foundation of trail development and human movement through Yellowstone. It is important to acknowledge the indigenous groups that traditionally lived on this land. In doing so, the sense of untouched nature that national parks around the world can evoke is further refuted in favor of a more complex narrative colored by colonialist forces (Mbaria and Ogada, 2016).

Coupled with the advent of tourism in the late 1800s, cultural conceptions shaped human utilization of the natural environment until the early 1900s introduced trail design and built construction standards (NPS). The early 1930s saw the development of the Yellowstone Grand Loop, 142 miles of concrete roadway cutting through the center of the park. With this mobility development, visitor traffic became a driving factor of various ecological and geographic shifts. Annual visitation in the park has marginally increased in the last decade, where rates increasing upwards of 40% and recreation visitations nearing 5 million (NPS) have resulted in unprecedented traveler and vehicle densities. Built development of trails and roadways, campsites and lodging facilities, and the construction of other user amenities that accounted for projected population growth have taken over large portions of previously natural land areas. Modern growth analyses and literature outlining Yellowstone's human presence have made

evident concerns regarding overpopulated backcountry campsites, heavy vehicle traffic, and limited space for lodging. Campsite expansion, roadway widening, and employee housing development are modern actions being carried out to minimize these issues, further building up the natural ecosystem.

Data

Name	*Geographic Unit*	*Source*	*Link*
Administrative Boundaries of National Park System	Vector Polygon	National Park Service	https://irma.nps.gov/DataStore/Reference/Profile/2225713?lnv=True
Elevation	Raster	USGS	https://apps.nationalmap.gov/downloader/
Developed Area	Vector Polygon	YELL Spatial Analysis Center (2021)	https://www.arcgis.com/home/item.html?id=e5816efd8b1f4059866ff35e553b0b98
Conservation Area	Vector Polygon	Yellowstone Center for Resources (2021)	https://www.arcgis.com/home/item.html?id=bad5f1fc95e242159291353f8bfaaa68
National Park Roads	Vector Line	National Park Service (2022)	https://www.arcgis.com/apps/mapviewer/index.html?layers=4042460c95c340f69a0b65a736a35a0a
National Park Trails	Vector Line	National Park Service (2022)	https://ufl.maps.arcgis.com/home/item.html?id=0086120c2bda4f929a931147a4c6f542
National Park Service Buildings	Vector Polygons	National Park Service (2019)	https://www.arcgis.com/home/item.html?id=2bf22e15d269431e9e3380382c0a1546
National Park Service Parking Areas	Vector Polygons	National Park Service (2019)	https://www.arcgis.com/home/item.html?id=685e6ac43db44a93915b8622034856e4
National Park Service Points of Interest	Vector Points	National Park Service (2022)	https://ufl.maps.arcgis.com/home/item.html?id=f437e40dbb3b4fbcb711842a5764aef4

Methodology

The following methodology began by selecting the Yellowstone National Park boundary from all National Park System Administrative Boundaries. All layers were projected in WGS 1984 Web Mercator (auxiliary sphere).

1. Manual Selection: Yellowstone
2. Export into New Layer “YNPBound”

Roads

To understand where the roads in the park are and the delays that can occur, the following methodology was applied. The road layer was clipped to the Yellowstone boundary. The roads were analyzed to classify the level of service which is a qualitative measure to compare the quality of traffic service. The level of service was determined by a transportation and mobility study conducted in 2016.

1. Select by Attribute: Select the segments of roads based off the study classified as D
    a. Export to New Layer “LevelofServiceD”
2. Select by Attribute: Select the segments of roads based off the study classified as C
    a. Export to New Layer “LevelofServiceC”
3. Select by Attribute: Select the segments of roads based off the study classified as B
    a. Export to New Layer “LevelofServiceB”

A 400 m buffer was implemented to give a general idea of areas impacted by automobile pollution. The size of the buffer was chosen from the Health Effects Institute's Special Report on Air Pollution (HEI, 2010).

1. Run buffer analysis tool
    a. Input = Roads
    b. Output = Road Buffer
    c. Distance = 400 Meters

Visitors & Human Density

In 2021, 4,860,242 people visited the national park. There are 5 entrances that could be used depending on the sites the tourists want to visit. A vector point feature class was created to show where the entrances are.

1. Clip the National Park Service Points of Interest to the Yellowstone boundary
2. Export into New Layer "YNP_POI"
3. Select by Attribute
    a. POINAME is equal to North Entrance
    b. POINAME is equal to Northeast Entrance
    c. POINAME is equal to East Entrance
    d. POINAME is equal to South Entrance
    e. POINAME is equal to West Entrance

Understanding where people enter the park, aids officials in directing funds to the correct area for infrastructure improvements. Using the entrance layer, a proportional symbology map was created on top of the Yellowstone boundary.

1. Add Field to Entrances attribute table (name: visitors)
2. Add number of visitors who used each entrance to attribute table
3. Edit symbology
    a. Primary symbology: Proportional symbols
    b. Field: Visitors
4. Add entrance point layer on top of the density map

Trails

The following methodology was used to analyze human impact through the use of recreational hiking trails, an important feature of the National Park. First, trails were selected by their surface and used to categorize meaningful line vectors into two layers based on the tread (surface) of the trail. 'Earth' or 'Native' tread incorporates existing materials, while non-native tread imports materials such as asphalt and wood for the trail surface.

1. Use Select by Location within the Yellowstone Boundary on the NPS Trail Layer
    a. Export into New Layer "YellTrail"
2. Select by Attribute: Trail Surface = 'Earth'
3. Export into New Layer "EarthTrail"
4. Select by Attribute: Trail Surface = 'Asphalt', 'Concrete' OR 'Wood'; AND Trail Use includes 'Hiker/Pedestrian'

5. Export into New Layer “BltTrail”

Next, five online travel and hiking guides were parsed to find the fifteen most popular trails. The top five relevant results on Bing for the search term “Yellowstone National Park” were chosen. The sites’ high listings indicate their popularity with consumers.

1. Wildland Trekking (Wildland Trekking Team, 2022)
2. All Trails (Alltrails, 2022)
3. Trip Advisor (Tripadvisor, 2022)
4. Backpacker Goods (Backpacker Goods Team, 2021)
5. As We Travel (Benny, 2022)

Multiple websites were used to avoid skewed data and ensure general popularity is understood. The trails most frequently mentioned were considered the most popular. These trails naturally have the most visitor traffic and human impact. Results were indicated in the attributes table.

1. Add Field used to add ‘Popularity’ as a column in the attribute table.
2. Binary system used to indicate popularity. *Note: When selecting trails from the ArcGIS Pro, one trail was not included in the TSA analysis because it could not be found in attribute table’s listed trails*
    a. 1 = ‘popular’ trails
    b. 0 = all other trails
3. Select by Attribute
4. Export to new layer “PopTrails”

The below Trail-Slope Alignment (TSA) angle analysis was performed on all trails with a Native surface and the most popular trails with a Native surface. TSA describes how the trail's path matches with the terrain's contours.

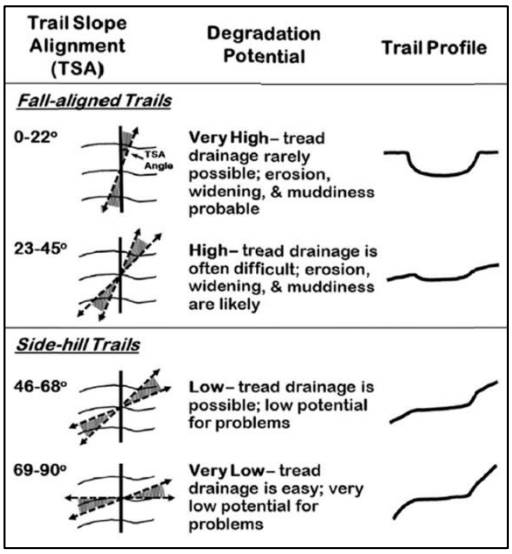


Figure 2 Explanation and Visualization of Trail Slope Alignment Angle Classes (Marion, et al., 2017)

The two TSA types that a trail can fall under are a 90-degree side-hill trail and a 0-degree fall-aligned trail. A side-hill trail follows the landscape's contour lines, while a fall-aligned trail follows the slope of the landscape perpendicular to contour lines. Side-hill trails allow for easier water drainage. Because the trail follows the contours of the landscape, it is more difficult to walk outside the established tread. This decreases trail widening, which protects natural resources.

On the other hand, fall-aligned trails have higher levels of erosion and widening both from humans and water. As water flows on the tread surface, it follows a 'fall line' that aligns closely with a trail's path when the TSA is low. Water will go down a trail, eroding the soil. It is also easier for people to walk outside the established trail because the areas to the direct left and right match the trail's slope and aspect. This contributes to trail widening, which kills vegetation and expands human impact from its intended bounds (Marion, et al., 2017).

To calculate TSA:

1. Import Elevation Layers.

   a. Use Mosaic to New Raster Tool to Merge USGS and Clip to the area of interest, Yellowstone National Park's boundaries. Name this new raster dataset "USGSElevMrg.tif"
      i. Pixel Type is 32 bit, Float; Bands = 1
   b. Use the Aspect Tool to create a new raster layer, 'YellAspect'
2. Select Trails by Attribute: Popularity = 1
3. Export into New Layer "PopTrail". Distinguish these from other trails with increased thickness of the line symbology.
4. Use the interactive Split Tool to split PopTrail lines at their vertices.
5. Use the Add Surface Information Tool (ESRI, 2021) to add the mean Aspect for each segment into a new Field. It will automatically have the name Z_Mean, renamed to 'AspectAngle' for clarity.
6. Use the Calculate Geometry Attributes Tool (ESRI, 2022) to determine each trail segment's angle in relation to due North (Line Bearing) in a new Field 'TrailAngle' .
7. Create a new Field 'TSA' that is calculated as Surface Aspect - Trail Direction.
   a. Right click on the TSA column in the attributes table and input the following in the Calculate Field pop-up:

```
TSA = ifelse(!TrailAngle!,!AspectAngle!)
Code Block:
def ifelse(TrailAngle, AspectAngle):
OG_TSA = abs(TrailAngle - AspectAngle)
     if OG_TSA > 270:
          return abs(OG_TSA - 360)
     elif OG_TSA > 90:
          return abs(OG_TSA - 180)
```

```
else:
    return OG_TSA
```

8. Display symbology with four color classes, indicating the four classes of Trail Slope Alignment angles:
    a. Poinsettia Red: 0-22

       These are the trail segment at a critical level, meaning the highest potential for problems, as they are the closest to being completely fall-aligned.
    b. Orange Dust: 23-45
    c. Sage Dust: 46-68
    d. Peacock Green: 69-90

       These are the trails with lowest potential for problems, as they are the closest to being side hill trails.

Development & Built Environment

Human presence in the 2.2 million acres of Yellowstone National Park (YNP) has resulted in the built development of natural areas. To analyze the extent of development in the park, a vector polygon feature layer of the YNP boundary was added to the map. A vector polygon layer of total developed area was then juxtaposed on top of the park boundary. Values were rounded to the nearest whole number; decimal places were set to zero when calculating geometry and statistics.

1. Add Field to DevArea attribute table (name: area)
2. Calculate Geometry (field: area), (unit: acres)
3. Summary Statistics (field: area), (statistic type: sum)

There are thousands of built structures in YNP including lodging, service, and recreation facilities. Because of climate conditions in the area, not all facilities are open for human use year-round. To determine what areas of the park were not exposed to human activity year-round, the NPS Building polygons were first selected within the YNP boundary.

1. Select by Location
    a. “NPSBuildings” intersect “YNPBound”
2. Export into New Layer “YNPBuild”
3. Select by Attribute
    a. “YNPBuild” where SEASONAL is equal to YES
4. Export into New Layer “Build_szn”
5. Run Buffer on “Build_szn”
    a. 1610 meters (linear unit)
    b. New Layer created “BuildSZNBuf”
6. Dissolve “BuildSZNBuf”
    a. New Layer created “BuildSZN”

Not all areas of the park experience seasonal closures. To identify what developed areas of the park are used year-round, and therefore may be subject to elevated human impact, a Kernel Density analysis was conducted after vector polygons were rasterized. This was done with the NPS Building layer selected within the YNP boundary.

1. Select by Attribute:
    a. SEASONAL is equal to No
2. Export into New Layer “Build_yr”
3. Run Buffer on “Build_yr”

a. 1610 meters (linear unit)

b. New Layer created “BuildYR”

4. Run Feature to Point on “BuildYR”

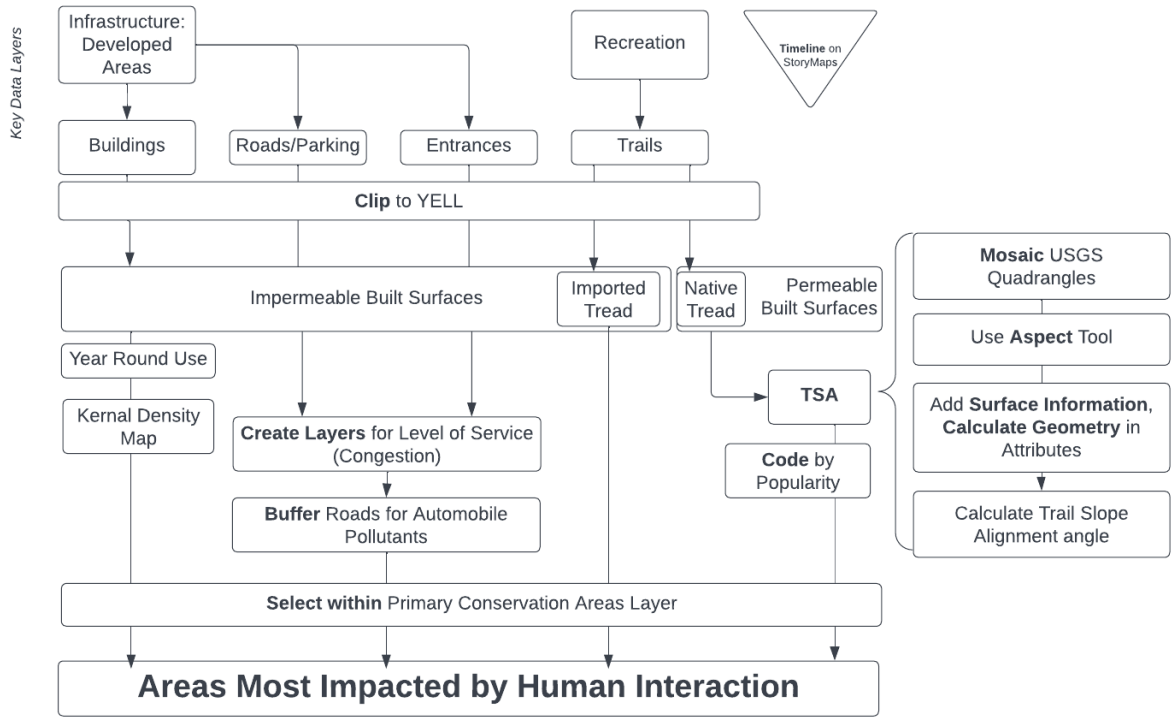
   a. New Layer created “YR_FTP”

5. Run Kernel Density Analysis on “YR_FTP”

   a. New Layer created “YR_KERN”

6. Change symbology (method: quantile) (color scheme: plasma)

Methodology Flowchart



Results

Roads

There are approximately 252 miles of roads in the park that all connect to allow people to enter and exit where they desire. Some segments of road are closed periodically throughout the year due to weather and infrastructure projects. Yellowstone National Park serves millions of visitors each year, with the number of visitors comes concerns about pollution caused by vehicle emissions and the amount of time to travel through the park.

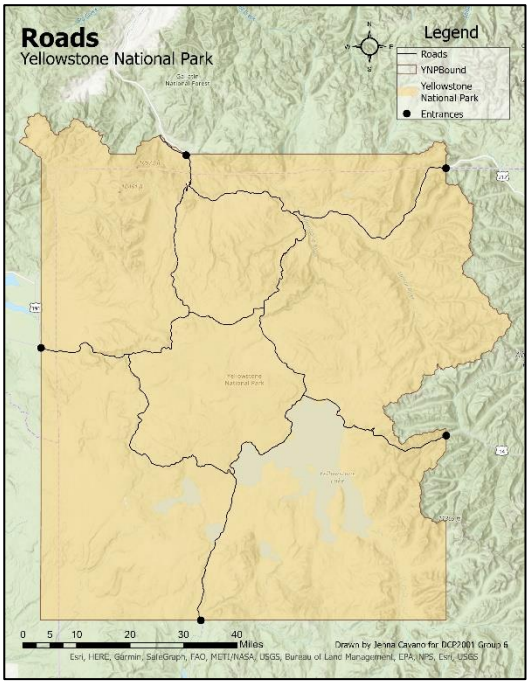


Figure 3: Roads in Yellowstone National Park

Level of Service	Description	Percent Time Spent Following
A	Speed would be controlled primarily by roadway conditions. A small amount of platooning would be expected.	< 40.0
B	The degree of platooning becomes noticeable. Some speed reductions are present.	> 40.0 to 55.0
C	Most vehicles are traveling in platoons. Speeds are noticeably curtailed.	> 55.0 to 70.0
D	Platooning increases significantly. Passing demand is high, but passing capacity reaches zero. A high percentage of vehicles are now traveling in platoons.	> 70.0 to 85.0
E	Demand is approaching capacity. Passing is virtually impossible. Speeds are seriously curtailed. The lower limit of this LOS represents capacity.	> 85.0
F	Whenever demand flow in one direction or both directions exceed the capacity of the segment. Operating conditions are unstable, and heavy congestion exists.	----

Table 1 Roadway level of service standards

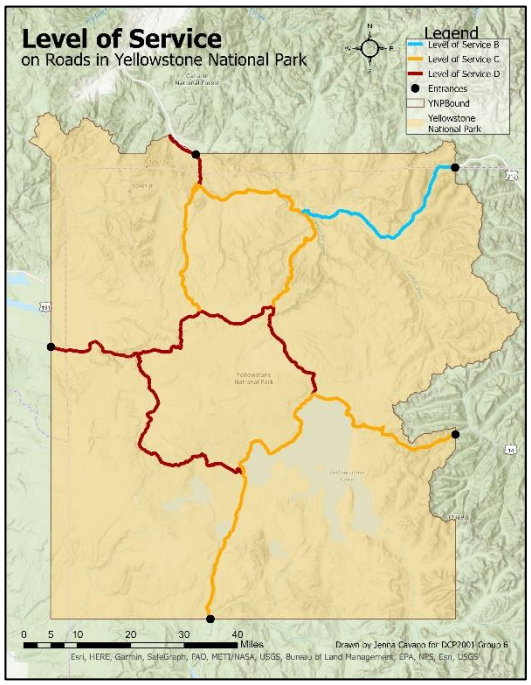


Figure 4: Level of service on roads

In 2016, a study was completed to understand the traffic and roadway conditions in Yellowstone National Park (Atwell et al., 2017). The level of service (LOS) analysis categorizes traffic flow and assigns quality levels based on performance measures such as speed and density. There are six levels of service that the roads could be classified as ranging from A to F. LOS A represents free flowing traffic while LOS F represents an excessive delay or a breakdown in flow. As a result of the study, the roadways in the park were classified as either B, C, or D. This means that visitors would spend 40% - 85% of their time on the road following the car in front of them. The amount of time spent following increases from A to F.

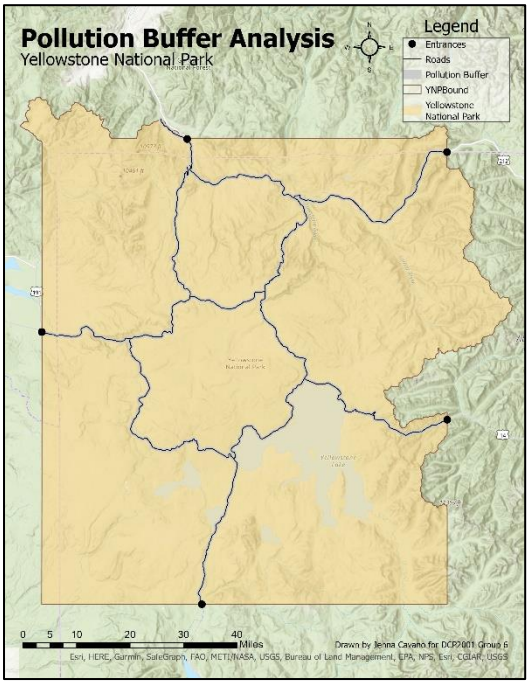


Figure 5: 400-meter buffer showing potential areas of impact from pollution

Automobile pollution is an issue that many locations must face but it is especially a concern in national parks as it impacts the visitor experience, vegetation, and wildlife. In 2021, the park allowed 2,976,625 vehicles to enter the park (*Annual Traffic Counts by Month*, 2022). The Health Effects Institute determined that the zones most impacted by traffic-related pollution are up to 300 to 500 meters from major roads (Institute, 2010). A 400-meter buffer was applied to the roads in Yellowstone to determine the areas where automobile pollution could occur. The pollution impacts visitor's visibility of the monuments they are there to visit. It also can impact the natural resources on the land leading to unintended side effects.

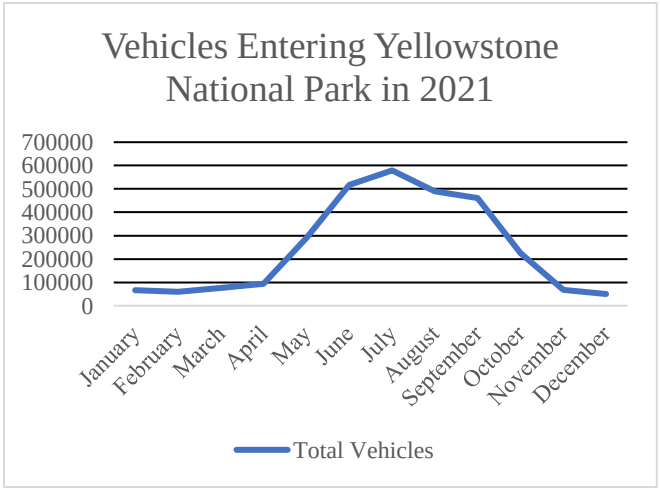


Figure 6: Vehicles entering the park in 2021

Visitors & Human Density

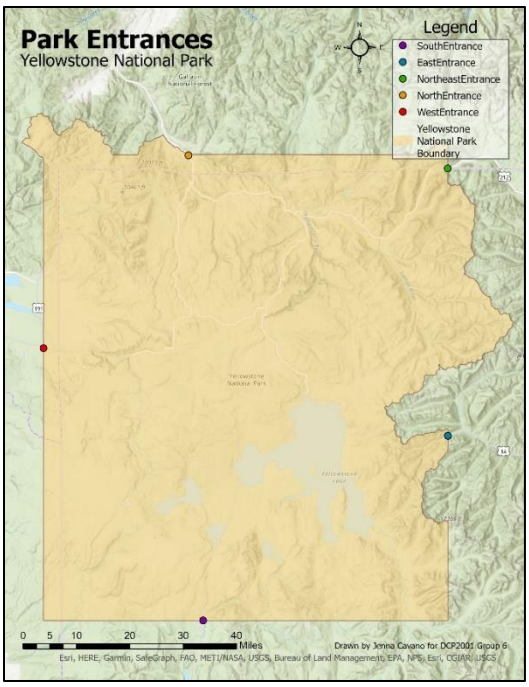


Figure 7: Entrances to Yellowstone National Park

There are five entrances to Yellowstone National Park: North, Northeast, East, South, and West. The most visitors come to the park in the summer months more specifically.

There are many small towns that have come from the creation of Yellowstone National Park. The town of West Yellowstone may be one of the most well-known, has existed since the early 1900s but was incorporated in 1966. The town is small but offers a variety of activities and places to stay for those who are looking to explore the national park during the day. Lodging in this town is the quickest way to visitors to see the geysers of Yellowstone. The proximity of the town to the park and the landmarks within the park has led to the west entrance being the most used entrance. As a result of the large number of visitors, there are increased delays on the section of road that the West entrance feeds into.

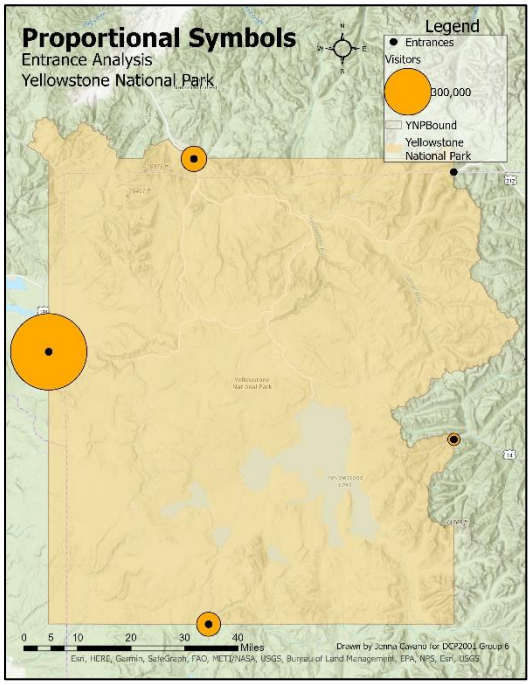


Figure 8: Entrance Analysis Showing Where Visitors Entered the Park in July 2021

Trails: The Trail-Slope Alignment Angle as a Predictor of Decay

Using the outlined methodology, the fourteen trails shown in Table 1 were chosen as the most popular trails, for which a Trail-Slope Alignment analysis is particularly significant. No trail was mentioned by all five websites analyzed.

Trail Name	Site 1	Site 2	Site 3	Site 4	Site 5	Total
Avalance Peak	☑	☑	☑	☐	☑	4
Mystic Falls	☐	☑	☑	☑	☑	4
Bunsen Peak	☑	☐	☑	☑	☐	3
Fairy Falls	☑	☑	☑	☐	☐	3
Lost Lake Trail	☐	☑	☑	☑	☐	3
Artist Point	☐	☑	☑	☐	☐	2
Hellroaring Creek	☑	☐	☐	☑	☐	2
Lone Star Geyser	☑	☐	☑	☐	☐	2
Mount Washburn	☑	☐	☐	☑	☐	2
North Rim Trail	☐	☑	☐	☐	☑	2
Observation Peak	☑	☐	☑	☐	☐	2
Osprey Falls	☑	☐	☑	☐	☐	2
Upper Geyser Basin & Obs Point	☐	☑	☐	☐	☑	2

Table 2: Popular Trails Identified for a Trail-Slope Alighnment analysis

Approximately 13% of all trail segments across Yellowstone's Earth/Native material trails were designed with a Trail-Slope Alignment angle that puts it at the highest risk of soil erosion. Data was cleaned by selecting for Trail Surface fields matching the description of Earth/Native. This is because non-native surfaces (boardwalks, paved trails, etc.) do not receive the same impact from high visitation.

For all trails, the mean TSA was 58 on a scale from 0 to 90. 13 percent of all trail segments analyzed were determined as being at a critical TSA level. A critical trail segment is

defined as having a TSA under 22, meaning that the trail is very close to being entirely fall-aligned. This acts as a predictor for the areas experiencing the most trail degradation.

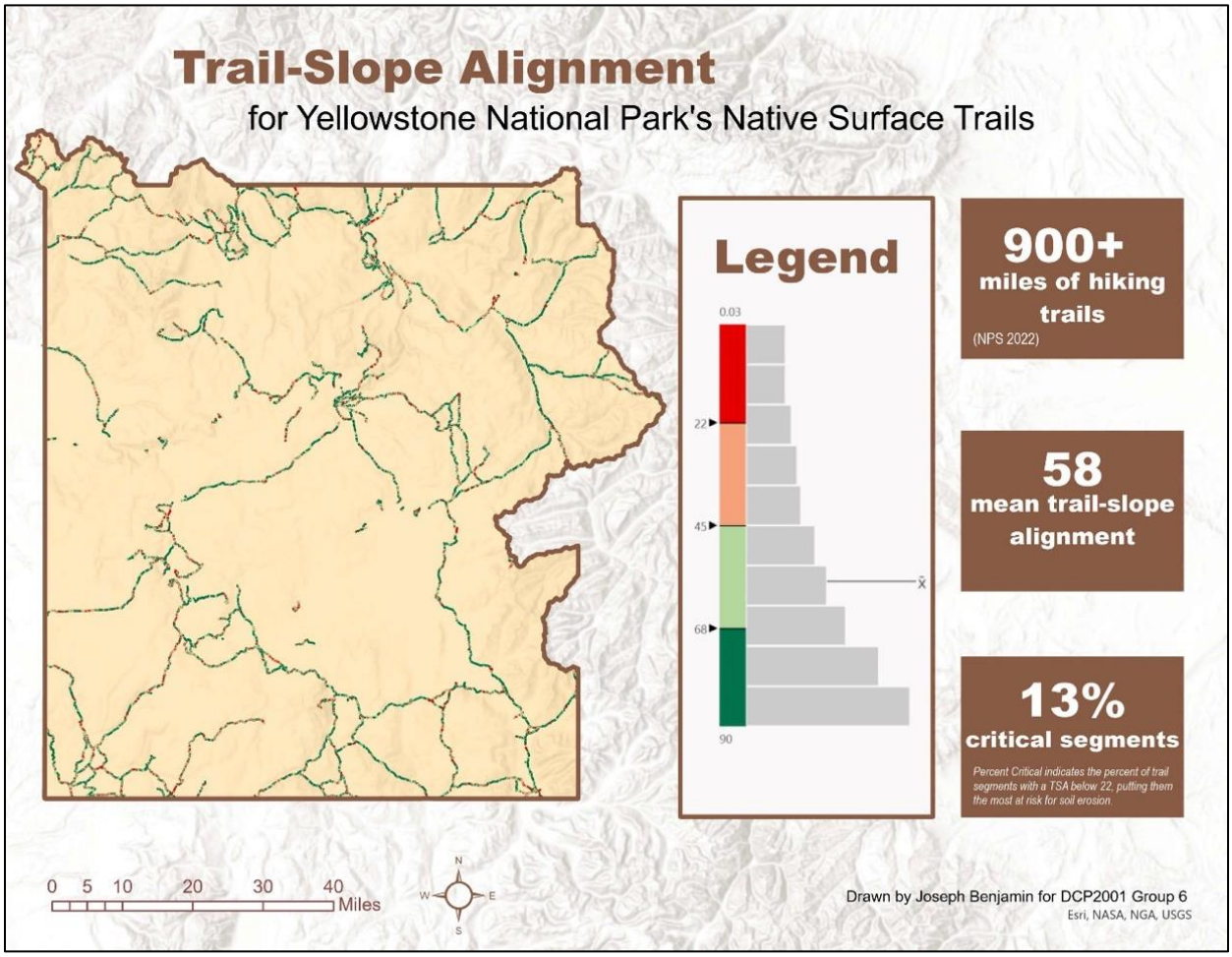


Figure 9: Trail Slope Alignment for Yellowstone National Park's Native Surface Trails

In addition to design, a highly trafficked trail will experience more intense human impacts, so Yellowstone's most popular trails were given extra consideration. The park's most popular trails are found in four clusters, with one outlier, the Avalanche Peak Trail. Of the 14 most popular trails analyzed in depth, nine of these had a higher proportion of trails at critical levels than the park's average of 13%. Three trails, the Avalanche Peak, Observation Point, and

Upper Geyser Basin Trails, had over 20% critical trail segments. This acts as a predictor of high levels of visitor impact in the form of trail erosion.

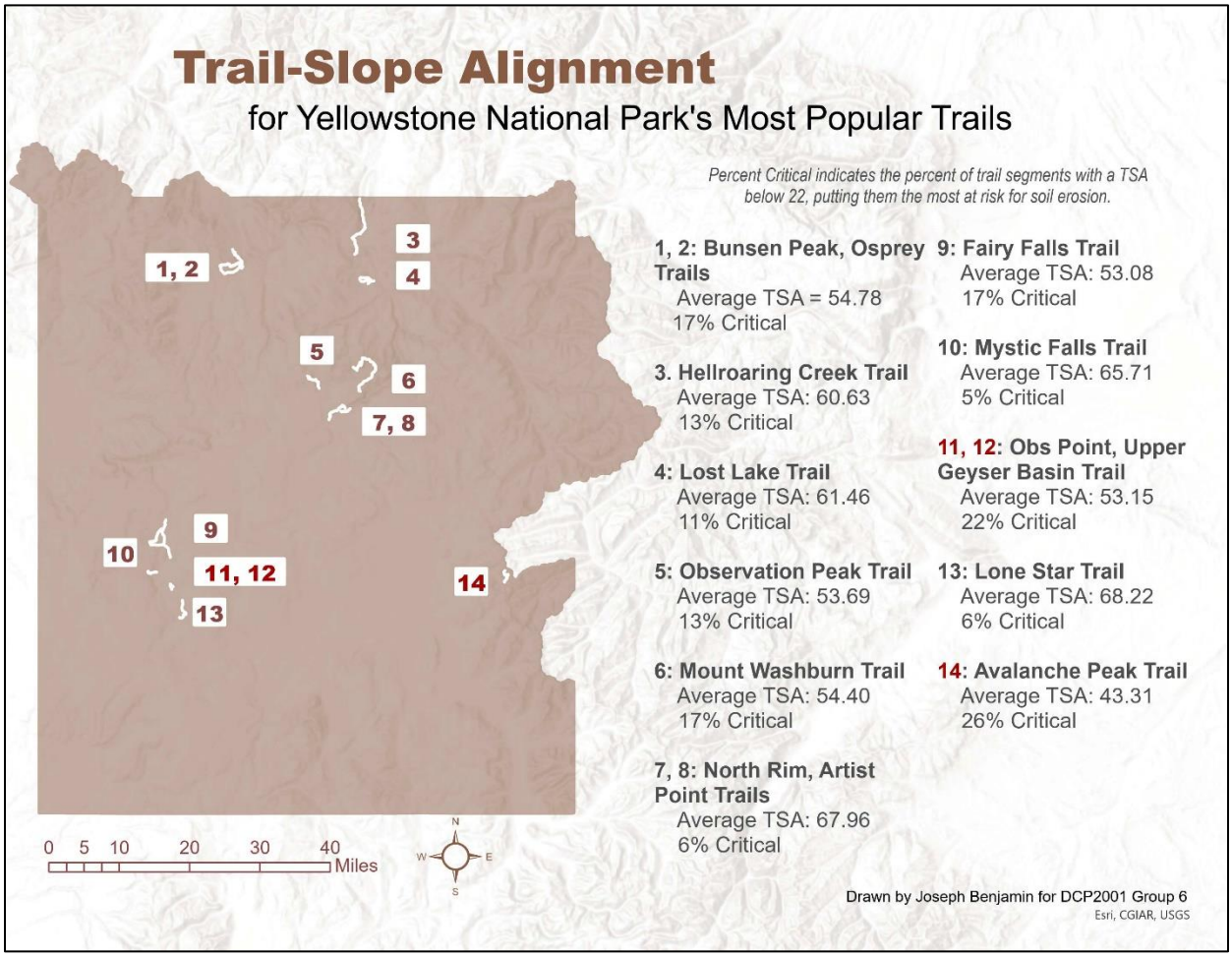
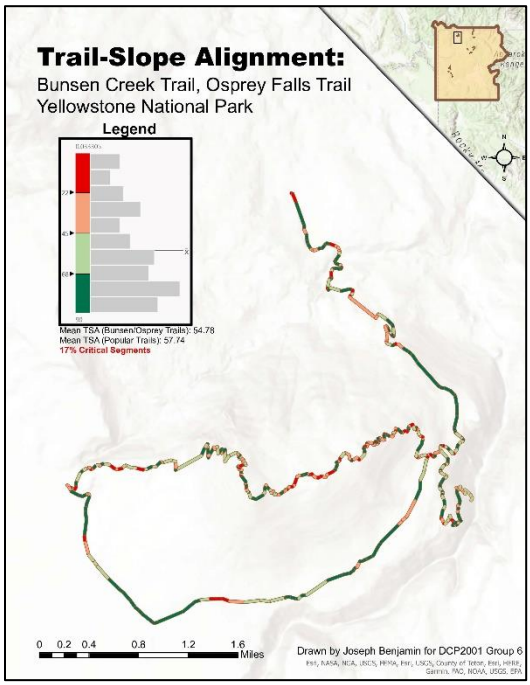


Figure 10: Trail-Slope Alignment for Yellowstone National Park's Most Popular Trails

Individual results for each popular trail vary greatly. The percent critical trail segments ranged between six and 26 percent. The analysis of each trail can be seen below:



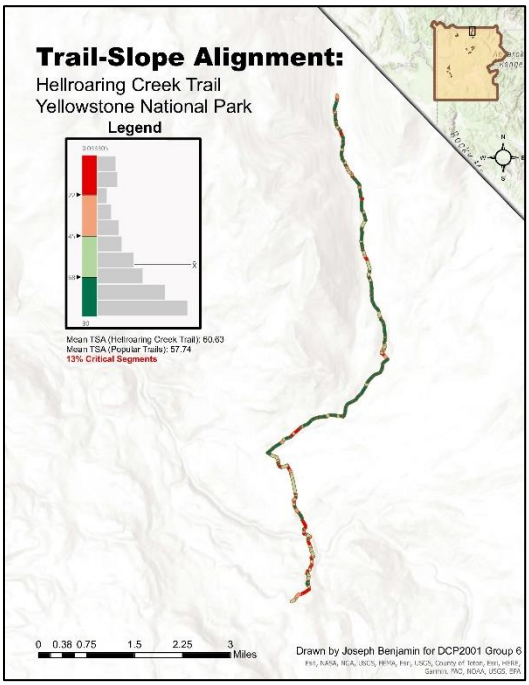
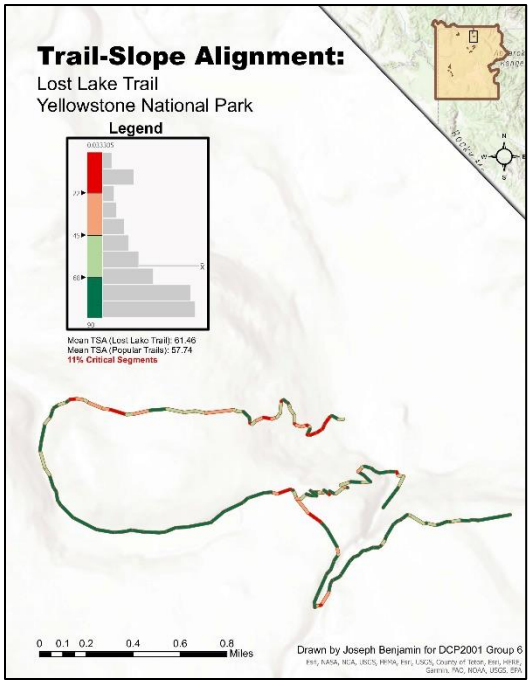


Figure 11: Left, Bunsen Creek and Osprey Falls Trails

Figure 12: Right, Hellroaring Creek Trail



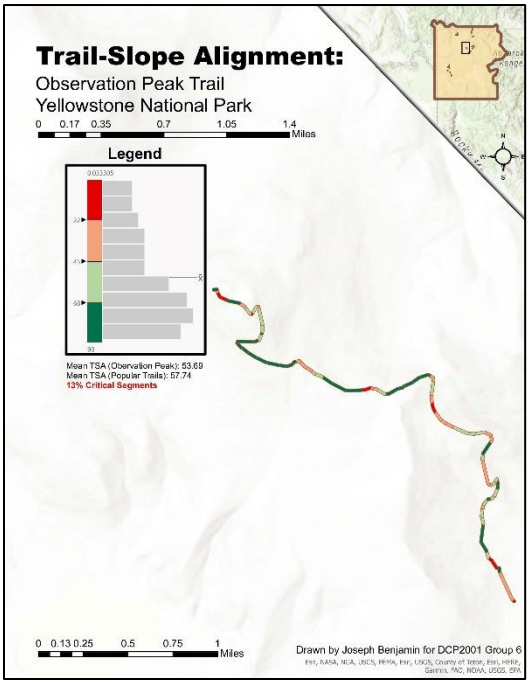
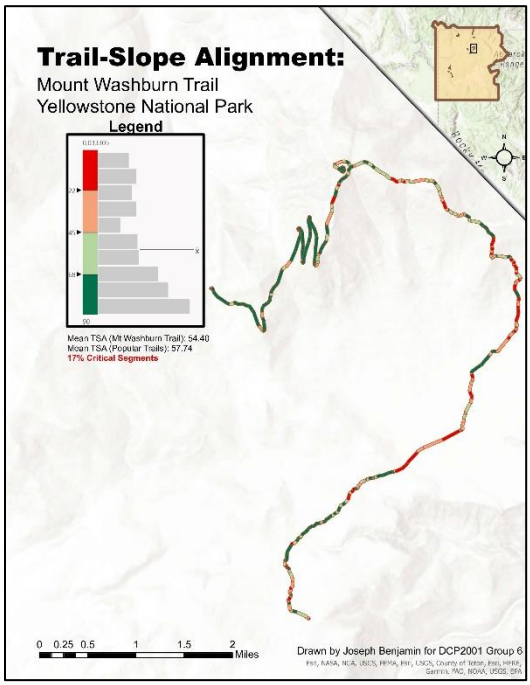


Figure 13: Left, Lost Lake Trail

Figure 14: Right, Observation Peak Trail



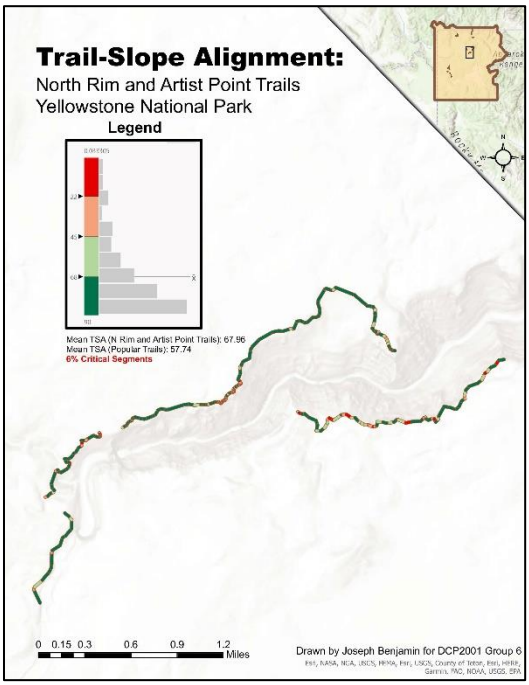
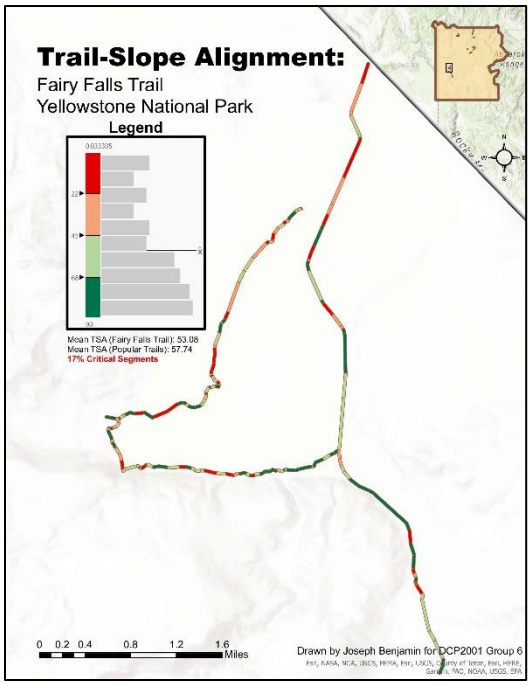


Figure 15: Left, Mount Washburn Trail

Figure 16: Right, North Rim and Artist Point Trails



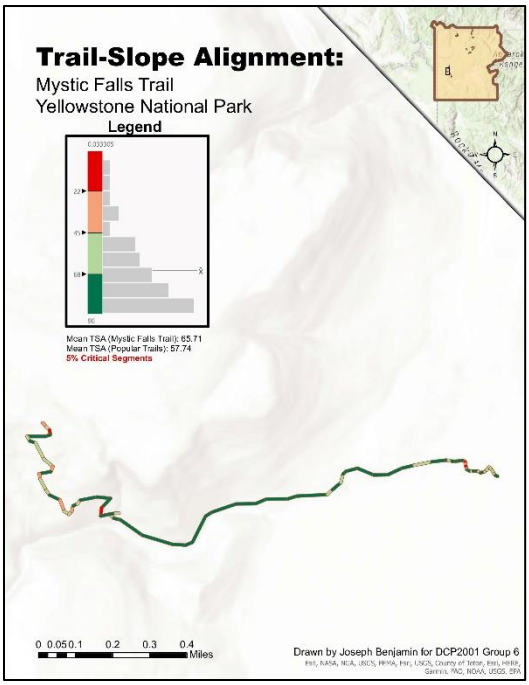
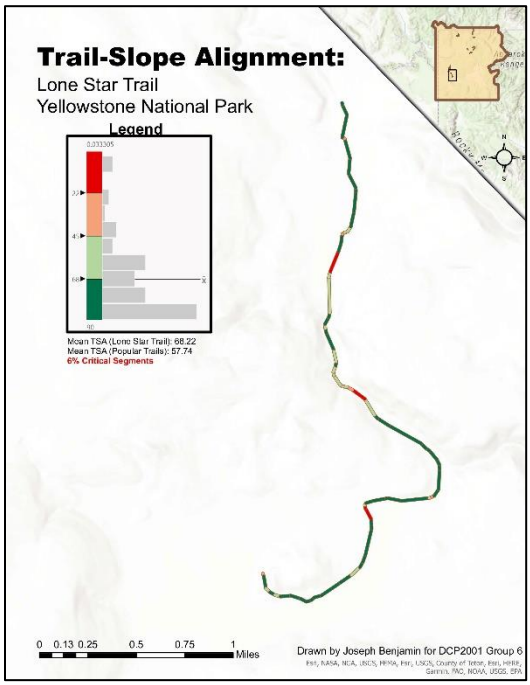


Figure 17: Left, Fairy Falls Trail

Figure 18: Right, Mystic Falls Trail



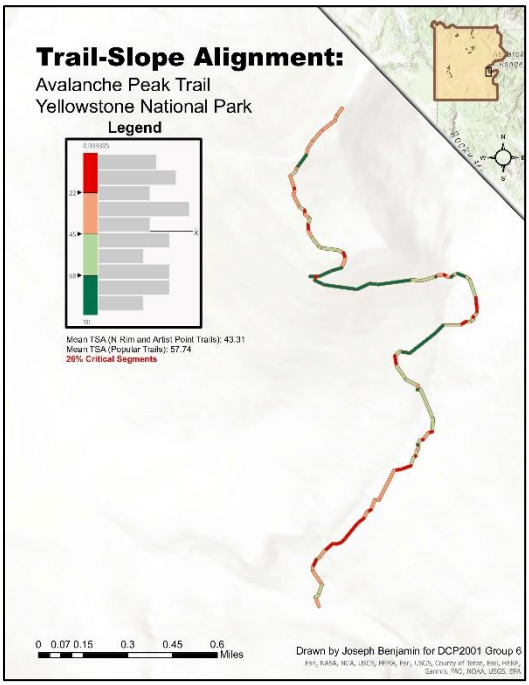


Figure 19: Left, Lone Star Trail

Figure 20: Right, Avalanche Peak Trail

It is important to note that other human impacts on the park through hiking trails was not captured by this analysis. For instance, the data collected strictly measures the official trail area and does not account for the creation of "social trails", the foot-worn paths that visitors unofficially creation by disturbing the vegetation in the area (Schmitt, 2016).

Development & Built Environment

Over the years as Yellowstone National Park (YNP) gained traction as a major tourism destination, parts of the park were built up to account for visitor density. Total developed area was calculated to be 4,600 acres, less than 1% of the park's total land area. Although miniscule compared to the total 2.2-million-acre area of the park, development of lodging and amenities in national parks and the impact on surrounding wild ecosystems remain a contended issue.

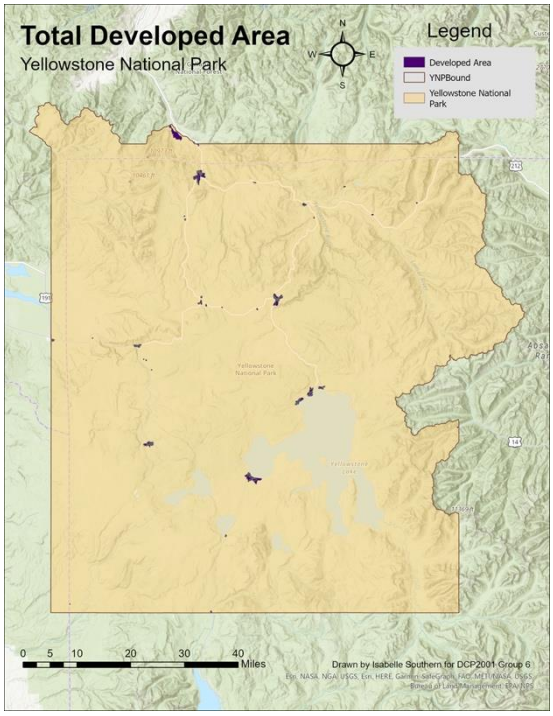


Figure 21: Total Developed Area

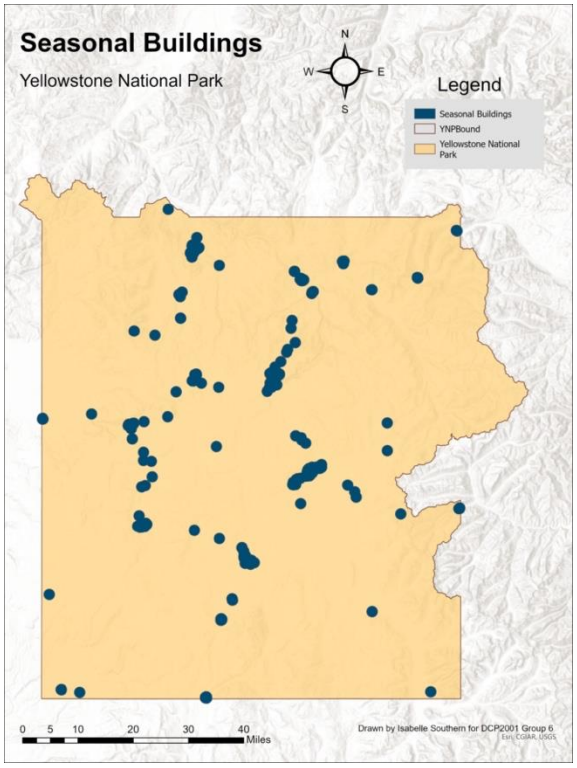


Figure 22: Seasonal Buildings

Due to harsh winter weather, many buildings in YNP operate on a seasonal schedule. Several lodging, amenities, and entertainment facilities do not experience human traffic for a large portion of the year. These built areas remain in the park, however intermittent use results in areas less impacted by visitor density. Data counts include 1,109 buildings operating seasonally, closed for the winter season, ranging from September to June.

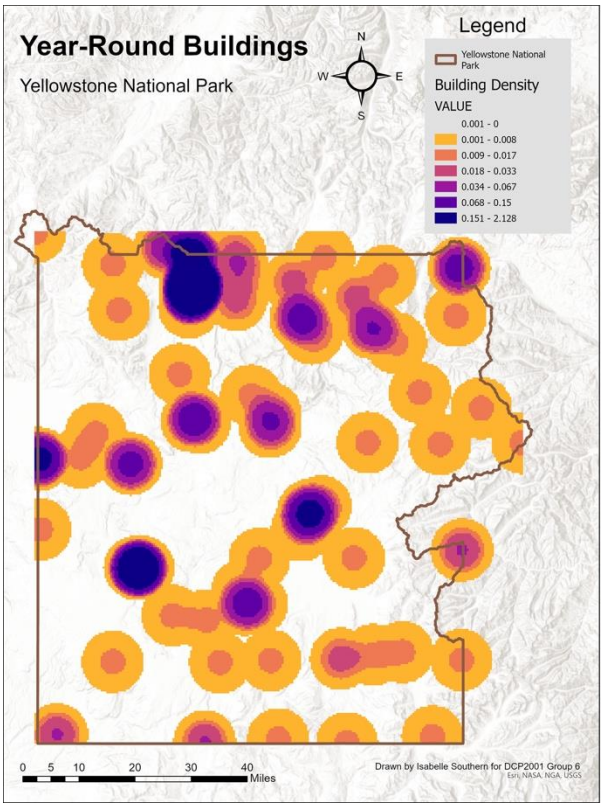


Figure 23: Year-Round Building Usage

Kernel Density Analysis was conducted to visualize the distribution of built areas used year-round, experiencing more intense human impact. The result was a heat map delineating the concentration of buildings not experiencing seasonal closures. Densely populated trends are apparent in the north section of Grand Loop Rd. Identifying areas of most use aids decisions about the allocation of services like wildlife conservation, built environment maintenance, and population control.

Conclusions

This research drew upon four indicators of human impact on the Yellowstone National Park ecology; roads, trails, built development, and visitor density. Juxtaposing the existing spatial data reflecting these four indicators aids in future planning and maintenance programs by identifying any changes or irregularities in the park's natural ecological condition.

Roadways and trails are indicators of human movement throughout the park. There are five entrances and approximately 252 miles of road in Yellowstone National Park. Most people choose to enter through the West entrance as that is the simplest way to view all the attractions though the roads within the park all connect. Vehicle pollution is a concern due to the large

number of vehicles that are permitted to enter the park. As a result of all the vehicles and long following times in certain sections of the park, visibility is occasionally limited due to a haze from the pollution. Yellowstone sees most of its visitors in the summer months as that is when a significant number of roads are open. Visitation has consistently peaked in July. Of the most popular trails, the Avalanche Peak, Observation Point, and Upper Geyser Basin Trails had critical trail segment levels that were almost double the average for Yellowstone trails, meaning the highest potential for problems, as they are the closest to being completely fall-aligned. These trails' popularity compounds the issues of poor TSA design, like soil degradation and trail widening. Further areas of research for trail use, maintenance, and design include documenting social trails & integrating other variables like trail grade into a sustainability analysis for the park's trail system.

Calculated as less than 1% of total park area, the presence of developed area in YNP does not account for a significant contribution to overall human impact. However, it is the distribution of buildings that harbor visitors year-round in areas of delicate ecology that contributes to environment-altering impact. For example, the development of lodging in Mammoth Hot Springs near the north entrance and the development of the town of West Yellowstone have increased mobility and recreation availability for tourists. Back country wilderness further away from main roadways in YNP is less susceptible to human traffic, however built development of lodging and recreation facilities is concentrated along delicate ecosystems like those of the Yellowstone River at Canyon Village, and the geysers near West Yellowstone.

Notable areas of impact include Mammoth Hot Springs in the north and the geyser basins in the west end of the park. Not only are these areas among the most fragile in terms of natural ecosystem function, these areas are very well connected by roads, trails, and are centered in some of the main developed areas in the park.

Identifying areas and scales of human impact around Yellowstone National Park aids in land use decision making and maintenance planning. Wildlife, vegetation, hydrology, and other delicate natural systems are carefully monitored and protected by the National Park Service, and the introduction of unmanaged human population movement through the park can quickly alter these natural ecosystems forever. No matter how small human presence may seem to be, the interconnected systems allowing and encouraging human movement in the park have consequences on the wild character the park is intended to protect.

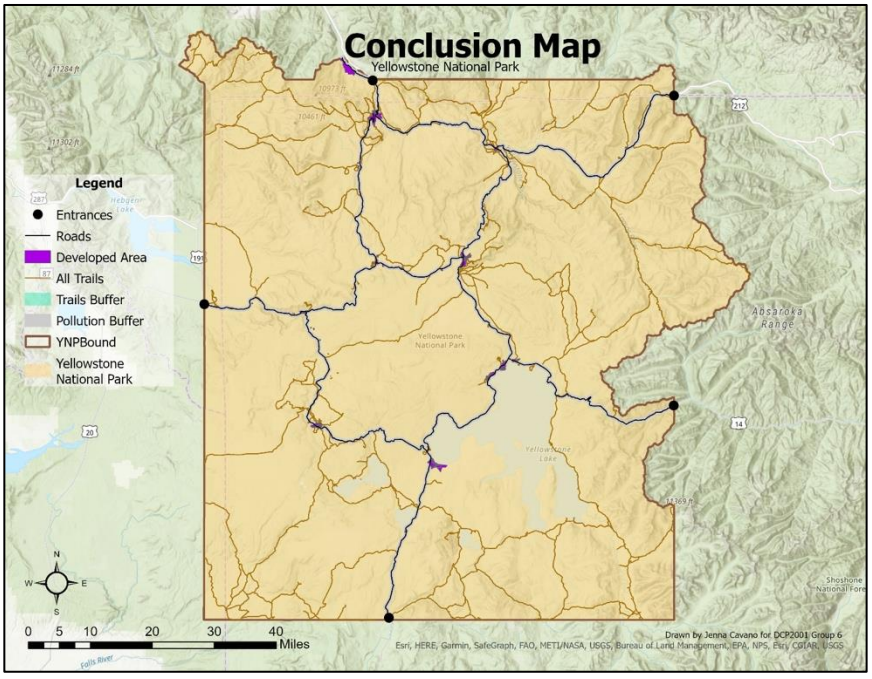


Figure 24: Conclusion Map

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